C1 AIEMLOPS CAPSTONE

NYC Taxi Demand & Fare Forecasting

End-to-End MLOps: Data → Training → Serving → CI/CD →
Monitoring → Responsible AI → Dashboard

Student Identity

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October 2026

Capstone Phases Overview

1 — Data Pipeline
Ingest → Clean → Aggregate → Feature Engineering

2 — Model Training
Optuna HPO · LightGBM · MLflow Tracking & Registry

3 — Model Serving
FastAPI · Fare · Demand · Batch Forecast · SHAP Explain

4 — CI/CD & Docker
GitHub Actions · Lint · Tests · Docker Build & Run

5 — Monitoring & Drift
PSI · KS Test · DriftReport · Performance Baseline

6 — Responsible AI
SHAP Explainability · Fairness Analysis · Model Cards

7 — Dashboard
Streamlit · Live API · Drift Panel · Explainability UI

8 — Golden Path
run_all.sh · Ansible E2E Verification · Docker Deploy

Phase 1 & 2 — Data Pipeline & Model Training

Phase 1 · Data Pipeline

▸ **Ingest**	TLC Parquet + Zone CSV → data/raw/
▸ **Validate**	Schema checks, cleaning bounds, CleaningReport
▸ **Aggregate**	Dense hourly grid: 253 zones × 744 hours
▸ **Features**	Demand: lags 1/2/3/24/168h + rolling stats
▸	Fare: distance, duration, zones, calendar
▸ **Split**	80/10/10 chronological (no leakage)
▸ **Tests**	102 unit tests — all green

Phase 2 · Model Training

▸ **Algorithm**	LightGBM (beat XGBoost in both problems)
▸ **HPO**	Optuna TPE · 25 trials · val RMSE criterion
▸ **Tracking**	MLflow · params, metrics, artefacts logged
▸ **Registry**	taxi-demand-forecaster v1
▸	taxi-fare-predictor v1
▸ **Demand**	test R^2 = 0.971 RMSE = 8.59 pickups/hr
▸ **Fare**	test R^2 = 0.955 RMSE = $3.47
▸ **Explain**	SHAP summary plots + model cards generated

Phase 3 & 4 — Model Serving & CI/CD Infrastructure

Phase 3 · FastAPI Serving

`GET  /health`	Model status · history_hours · zones
`POST /predict/fare`	Single-trip fare prediction
`POST /predict/demand`	Zone-hour pickup count
`POST /forecast/demand`	Recursive multi-hour batch forecast
`POST /explain/fare`	SHAP feature contributions
`POST /explain/demand`	SHAP feature contributions
`GET  /metrics`	Prometheus instrumentation

Phase 4 · CI/CD & Docker

▸ **GitHub Actions**	.github/workflows/ci.yml
▸ **Job: lint**	ruff — zero warnings
▸ **Job: unit-test**	pytest · ≥75% coverage gate
▸ **Job: integ-test**	smoke tests · no AWS required
▸ **Job: docker-api**	Dockerfile build verified
▸ **Job: docker-dash**	Dockerfile.dashboard verified
▸ **Docker Image**	taxi-forecasting-api:latest
▸ **Container**	/health → status: ok · both models loaded

Phase 5 & 6 — Monitoring & Responsible AI

Phase 5 · Monitoring & Drift

▸ **Capture**	JSON prediction logs per request (firehose-ready)
▸ **PSI**	Population Stability Index — quantile-binned
▸ **KS Test**	Kolmogorov-Smirnov two-sample per feature
▸ **Alert**	drifted=True when PSI ≥ 0.2
▸ **Demand**	demand_drift_metrics.json — 2 features drifted
▸ **Fare**	fare_drift_metrics.json — 2 features drifted
▸ **Perf**	Live RMSE vs baseline · 20% tolerance gate

Phase 6 · Responsible AI

▸ **SHAP**	TreeExplainer — global + per-instance
▸ **Fairness**	5 subgroup dimensions per model
▸ **Dims**	Borough · Time-of-day · Weekend
▸	Airport zone · Passenger count
▸ **Flagged**	3 of 5 dimensions flagged (disparity > 1.5×)
▸ **Cards**	docs/model_cards/demand.md + fare.md
▸ **Artefacts**	SHAP PNGs · fairness JSON committed

Phase 7 & 8 — Dashboard & Golden Path

Phase 7 · Streamlit Dashboard

▸ **URL**	http://localhost:8501
▸ **Overview**	Project summary + model metrics
▸ **Demand**	Zone selector · 24h forecast chart
▸ **Fare**	Live fare estimate with inputs
▸ **Explain**	SHAP bar chart — per-instance contributions
▸ **Resp. AI**	Fairness report + disparity table
▸ **Monitoring**	Drift heatmap · KPIs · performance baseline

Phase 8 · Golden Path

▸ **Entry point**	./run_all.sh
▸ **Stages**	Pipeline → Train → Serve → Drift → RAI
▸ **Portable**	make_mlflow_portable.py → mlflow.container.db
▸ **Docker**	docker build → docker run → /health: ok
▸ **Ansible**	ansible-playbook verify_e2e.yml
▸ **Result**	All 8 phases: PASS
▸ **Repo**	github.com/saibalban2-tech/taxi-forecasting-student

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All 8 Phases Completed

Phase 1	Data Pipeline	Phase 5	Monitoring
1.4M records	Ingested	3 drift reports	✓
Phase 2	Model Training	Phase 6	Responsible AI
RMSE 3.27	in 95	SHAP + fairness	✓
Phase 3	Model Serving	Phase 7	Dashboard
5 endpoints	live	8501	live ✓
Phase 4	CI/CD & Docker	Phase 8	Golden Path
GitHub Actions	✓	4 steps	PASS ✓

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